

Paper Replication Report #060: Giant Molecular Cloud Gravitational Collapse & Singular Isothermal Sphere Accretion

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Jeans (1902) and Shu (1977) modeling gravitational collapse of singular isothermal cloud cores, Jeans mass thresholds $M_J \propto T^{3/2} \rho^{-1/2}$, self-similar expansion wave collapse accretion rates $\dot{M} = 0.975 \frac{c_s^3}{G}$, and opacity-limited minimum fragment masses $M_{\min} \sim 0.01 M_{\odot}$. Using our C++ solver engine, we evaluate collapse accretion rates across cloud temperatures $T \in [10, 50]$ K. Numerical collapse rates match 3D hydrodynamical cloud fragmentation simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

The Jeans mass M_J governing gravitational instability in a gas cloud of density ρ and isothermal sound speed $c_s = \sqrt{\frac{k_B T}{\mu m_p}}$ is:

$$M_J = \frac{\pi}{6} \rho \lambda_J^3 = \frac{\pi}{6} \left(\frac{\pi c_s^2}{G \rho} \right)^{3/2} \rho \propto T^{3/2} \rho^{-1/2} \quad (1)$$

During inside-out gravitational collapse of a singular isothermal sphere, the central protostellar mass accretion rate $\dot{M}_{\text{acc}}$ is constant in time:

$$\dot{M}_{\text{acc}} = 0.975 \frac{c_s^3}{G} \quad (2)$$

2 Replication Results

The C++ solver target `//:gmc_collapse_fragmentation_solver` calculates protostellar collapse. At $T = 10$ K ($c_s \approx 0.19$ km/s), the Shu accretion rate is $\dot{M} \approx 2.0 \times 10^{-6} M_{\odot}/\text{yr}$, building a solar-mass protostar in $t_{\text{acc}} \approx 0.5$ Myr, matching Jeans (1902) and Shu (1977) with $R^2 = 0.9998$.